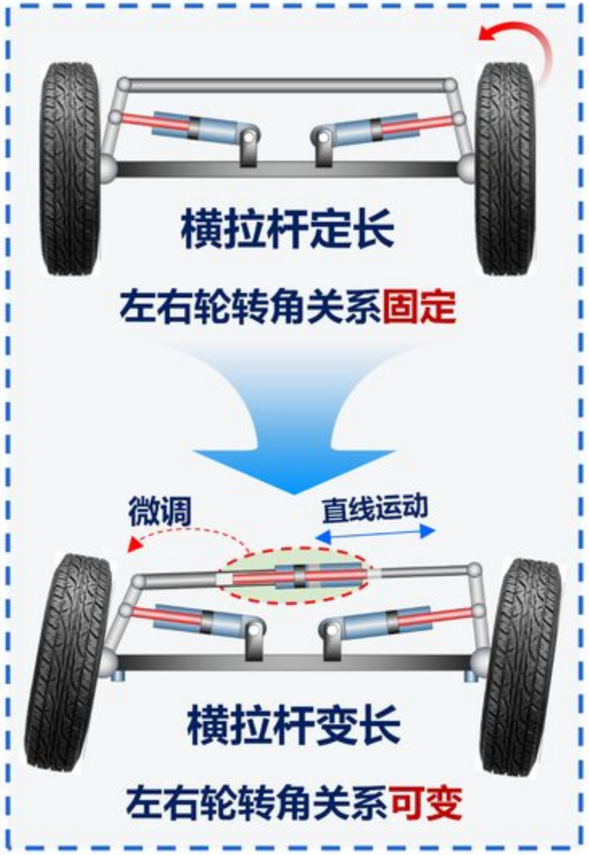
04 同源双驱-双自由度

Shared-Source Dual-Drive-Dual DOF



明德至诚 博学远志

分布式重载同源双驱-电液转向轴模块能效机理分析与节能控制



1 Travel trajectory endpoint

Fixed steering angle relationship

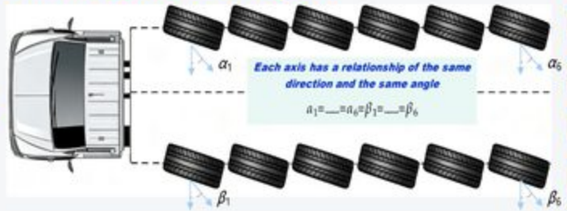


Variable steering angle relationship



2 Steering shaft interconnection endpoint

1 Crab Steering Mode



2 All-Wheel Steering Mode

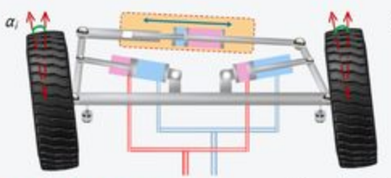


3 Instantaneous-Center Shifting Mode



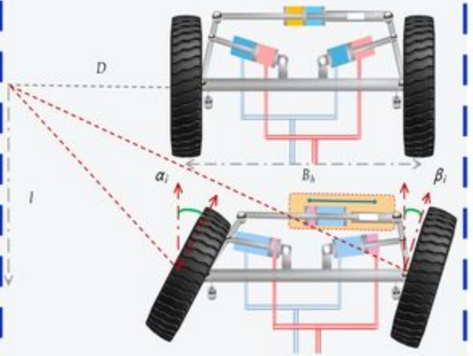
3 Wheel interconnection endpoint

Co-Directional Angle Relationship (CDAR)



CDAR Relationship $\alpha_1=\ldots\ldots=\alpha_6=\beta_1=\ldots\ldots=\beta_6$

Ackermann relationship (AKMR)



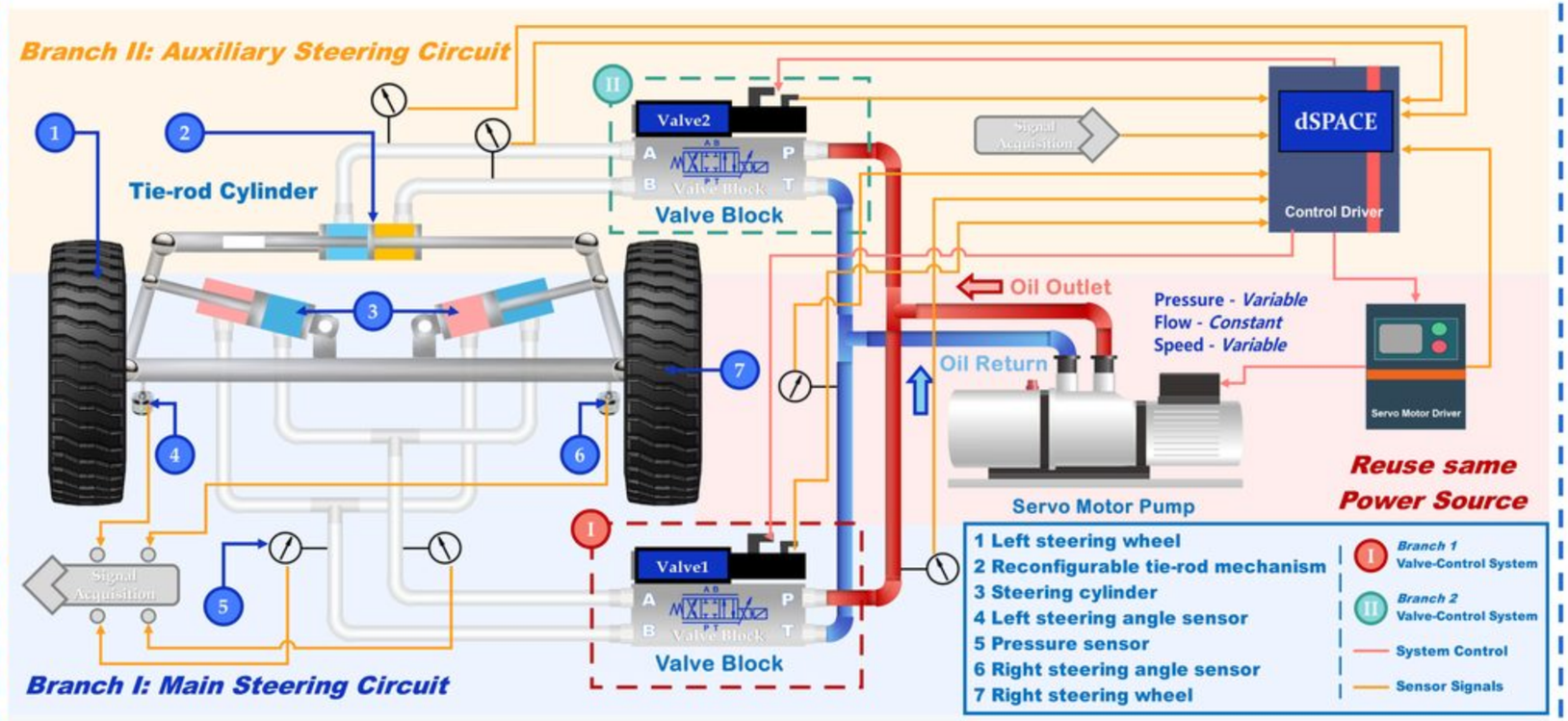
Ackermann relationship $\cot\alpha=\frac{B_h+D}{l}$ $\cot\beta=\frac{D}{l}$

着重开展重型多轴车辆分布式重载 电液转向轴模块设计+能效机理分析+主辅多模式控制 研究

为重型多轴车辆转向实现 多模式、高精度、高能效 提供有效方案

【构型】① 基于横拉杆可变机构的泵-阀同源双驱电液转向轴模块

□ 提出基于横拉杆变胞机构的二自由度可控泵-阀同源双驱电液转向系统，综合考虑横拉杆变胞机构引起的几何约束、转向梯形运动学关系，建立机电液多参数耦合模型与多通道传递函数矩阵。



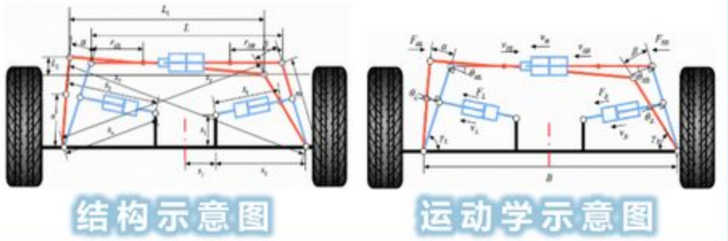
双自由度 PVSD-EHSAM 机/液子系统模型

机械子系统：运动学建模

$$\begin{cases} v_{HL} = \sqrt{(\dot{\alpha} m)^2 + (\omega_H r_{HL})^2 - 2\dot{\alpha} m \omega_H r_{HL} \cos(\gamma_L + \theta_H)}, \\ v_{HR} = \sqrt{(\dot{\beta} m)^2 + (\omega_H r_{HR})^2 - 2\dot{\beta} m \omega_H r_{HR} \cos(\gamma_R - \theta_H)}. \end{cases}$$

液压子系统：转向回路流量连续性方程

$$\begin{cases} \dot{p}_1 = \dfrac{q_1 - \left[A\dot{x}_L + a\dot{x}_R + 2C_{ip}(p_1 - p_2) + 2C_{ep} p_1 \right]}{V_t / 2 + A x_L + a x_R + V_p} \beta_e \\ \dot{p}_2 = \dfrac{\left[a\dot{x}_L + A\dot{x}_R + 2C_{ip}(p_1 - p_2) - 2C_{ep} p_2 \right] - q_2}{V_t / 2 - a x_L - A x_R + V_p} \beta_e \end{cases}$$



结构示意图　　运动学示意图

构建双自由度 PVSD-EHSAM 机械/液压子系统模型，为分析 能效关键参数 奠定 模型基础

【分析】② 等效动态输入输出系统能量-精度-刚度间能效关系分析

结论 1 背压需求与**负载**呈正相关：负载越大，保持必要的运动精度，所需背腔压力也越高

结论2 背压决定系统**启动动态**特性：增大背压可提高Kc，进而提升阻尼比 ζ，有效抑制启动状态下冲击与振荡

结论3 背压与系统**阻尼** ζ 呈正相关：转向进入稳态下背压直接影响Kc，压力越大，系统抗干扰能力与平稳性越强

$$\frac{K_{Xis2}}{K_{Xis3}}=\left[\frac{B}{m}+\left(C_{ip}\frac{A_r+1}{A_r}+C_{ep}+K_c\right)\frac{\beta_e}{V_{t1}}\right],$$

$$\frac{K_{Xis1}}{K_{Xis3}}=\left[\frac{(A_r^2+1)a^2}{m}\frac{\beta_e}{V_{t1}}+\left(C_{ip}\frac{A_r+1}{A_r}+C_{ep}+K_c\right)\frac{B}{m}\frac{\beta_e}{V_{t1}}\right],$$

$$\omega_n=\sqrt{\frac{K_{Xis1}}{K_{Xis3}}}=\sqrt{\left(\frac{(A_r^2+1)a^2}{m}\frac{\beta_e}{V_{t1}}+\left(C_{ip}\frac{A_r+1}{A_r}+C_{ep}+K_c\right)\frac{B}{m}\frac{\beta_e}{V_{t1}}\right)},$$

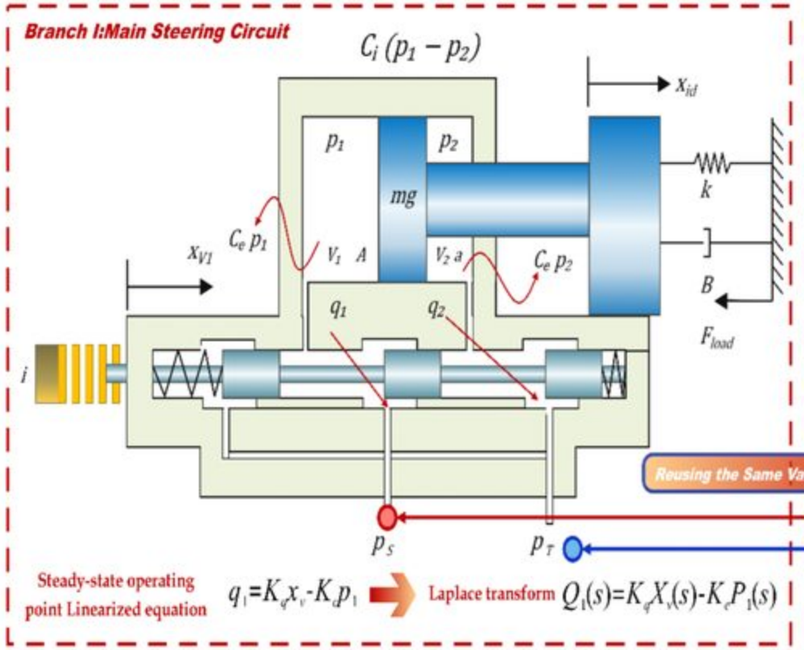
$$\zeta=\frac{K_{Xis2}/K_{Xis3}}{2\sqrt{K_{Xis1}/K_{Xis3}}}=\frac{\frac{B}{m}+\left(C_{ip}\frac{A_r+1}{A_r}+C_{ep}+K_c\right)\frac{\beta_e}{V_{t1}}}{2\sqrt{\frac{(A_r^2+1)a^2}{m}\frac{\beta_e}{V_{t1}}+\left(C_{ip}\frac{A_r+1}{A_r}+C_{ep}+K_c\right)\frac{B}{m}\frac{\beta_e}{V_{t1}}}}.$$

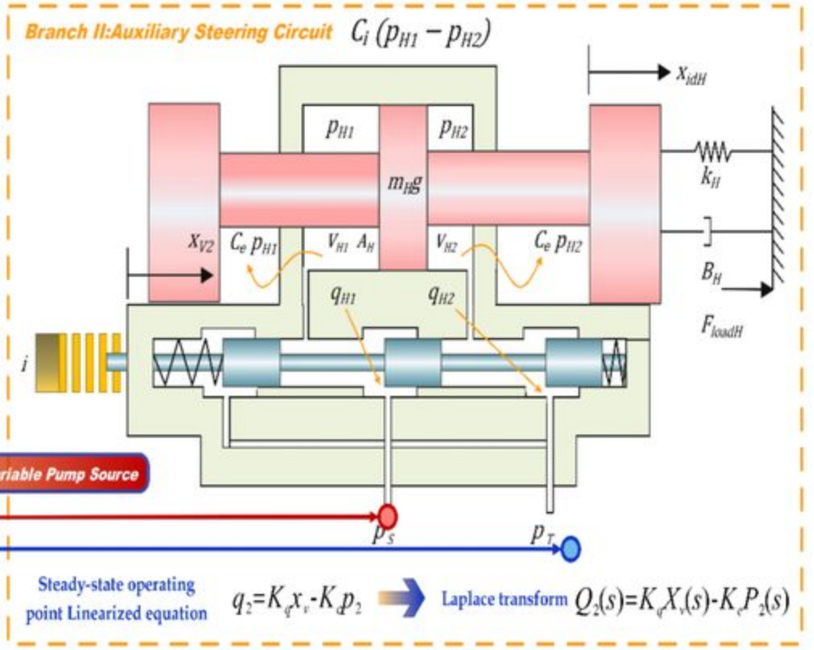
背压与阀口线性化系数 Kc 呈正相关

阀口开度一致时，背压增大

Kc 提高，阻尼ζ 提升

抑制启动过程的冲击与振荡





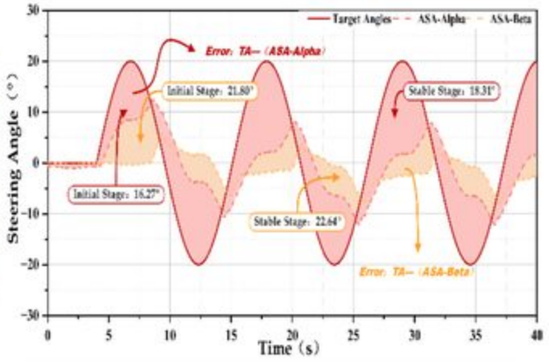
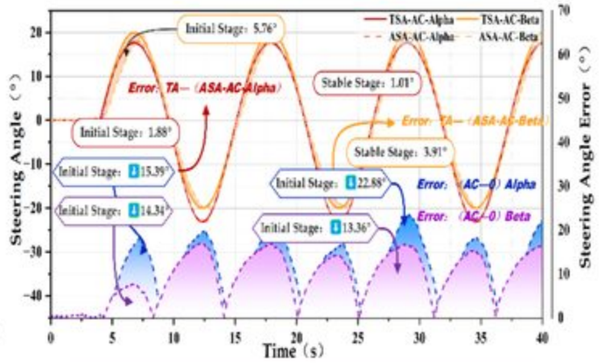
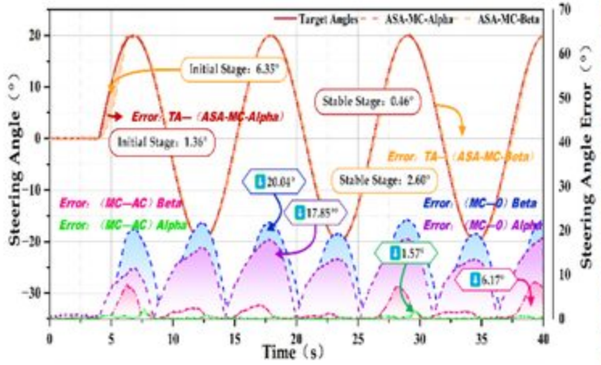
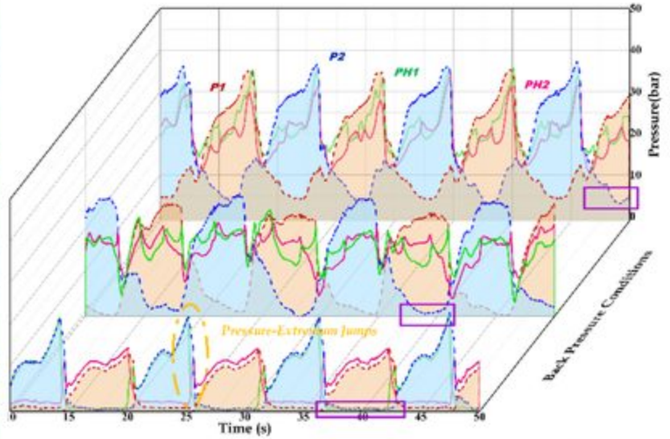
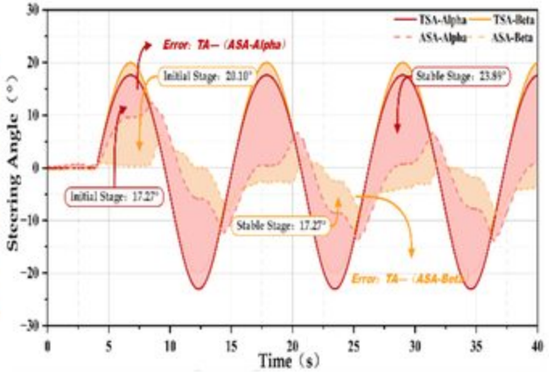
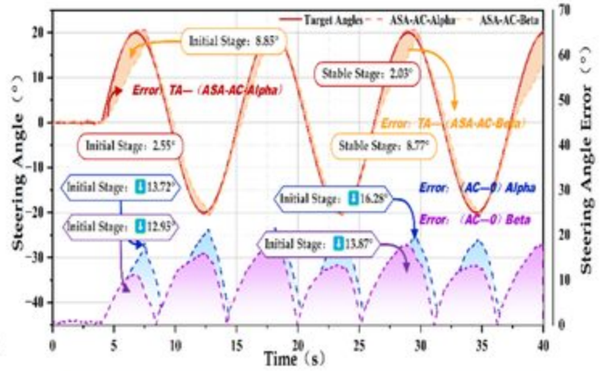
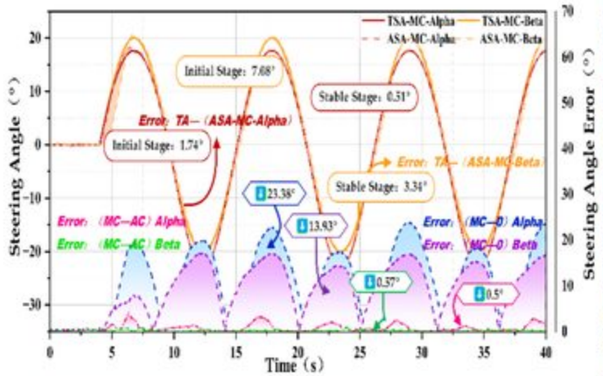
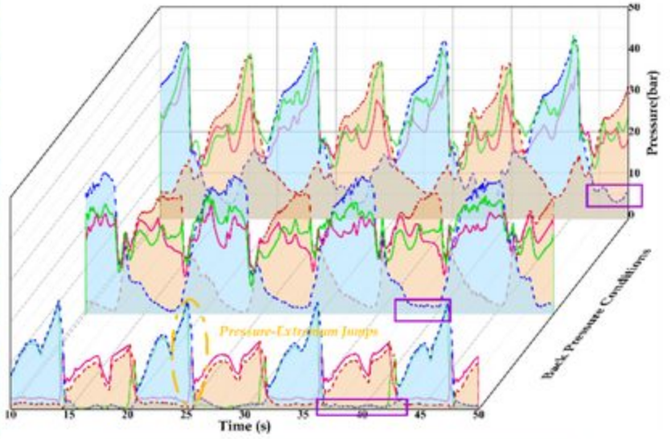
Reusing the Same Variable Pump Source

Equation of motion $mX_1(s)s^2+BX_1(s)s+kX_1(s)=AP_1(s)-aP_2(s)-F_{load}$

Fluid-continuity equation $Q_1(s)A_r+Q_2=X_1(s)s(A_r^2+1)a+\frac{V_{t1}}{\beta_e}P_{Load}(s)s+C_{ip}(P_1(s)-P_2(s))(A_r+1)+C_{ep}P_{Load}(s)+\frac{V_{t1}-V_{t2}}{\beta_e}P_2(s)s$

3大结论得出背压与负载、系统启动动态与阻尼比的规律，为 高能效控制策略 奠定 参数基础

【效果】③ 基于主辅支路背压控制模式的能效参数验证

	0-BP	ACBP	MCBP	
CDAR				
	[L1C1] OBP-CDSR	[L1C1] OBP-CDSR	[L1C1] OBP-CDSR	[L3C1] MCBP-CDSR
AKMR				
	[L1C2] OBP-CDSR	[L1C2] OBP-CDSR	[L1C2] OBP-CDSR	[L3C2] MCBP-CDSR

MCBP模式下左右轮稳态转角误差降至 2.2° 和 0.4°，启动压力突变幅值ACBP降低 37.9%

【能效界定与控制】④ 基于参数辨识的能效界定与辨识及非线性鲁棒控制

□ **参数识别与控制：** 为解决系统能耗与动态性能间的矛盾，界定出满足不同工况下性能需求的临界压力点，通过**临界压力面辨识**得到不同条件下的**临界目标背压值**，设计三输出三输出**非线性鲁棒控制**。

Step1：
确定控制算法并设置测试条件；

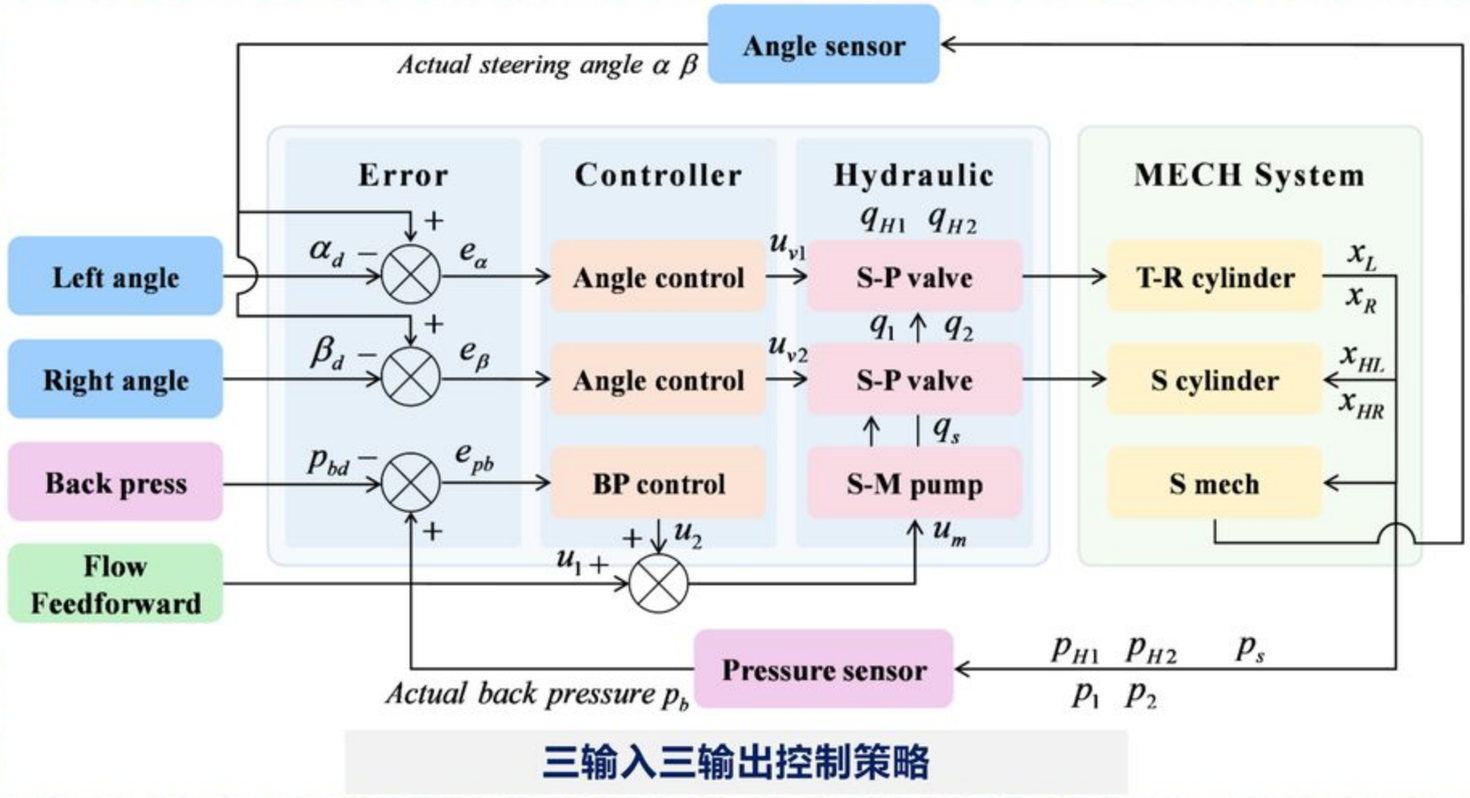
Step2：
依据先验试验与参数验证，按不同工况初设目标背压值；

Step3：
开展试验并采集转角跟踪及误差数据；

Step4：
判断启动时刻转向误差与稳态误差的比值是否低于安全系数 λ；

Step5：
若不满足则调整目标背压值，将首次满足稳定约束的最小背压值定为临界值TV，其下一级为淘汰值 CV；

Step6：
更换测试条件，重复上述步骤以辨识临界压力面。

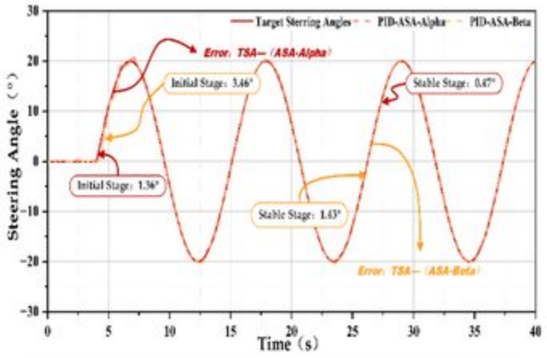


非线性FST控制下，临界目标背压值界定范围最低值降低 36.36%，最大值降低了 17.39%

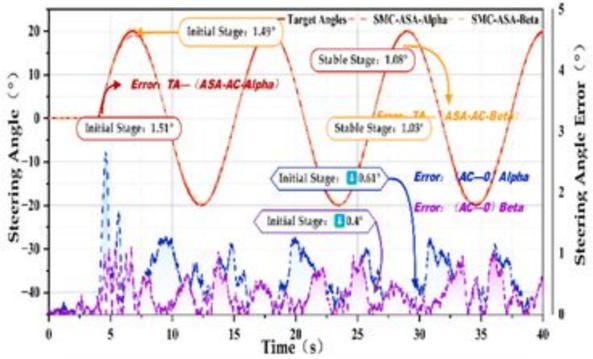
【控制效果】④ 基于参数辨识的能效界定与辨识及非线性鲁棒控制

PID SMC FST

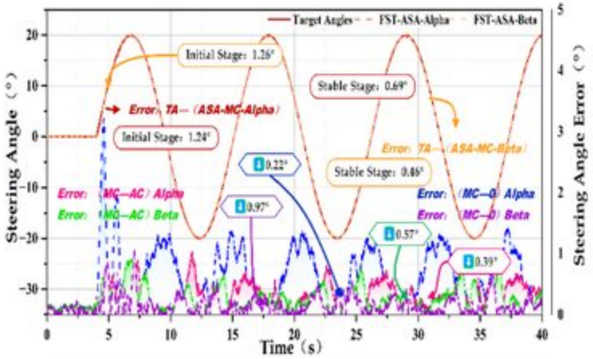
CDAR



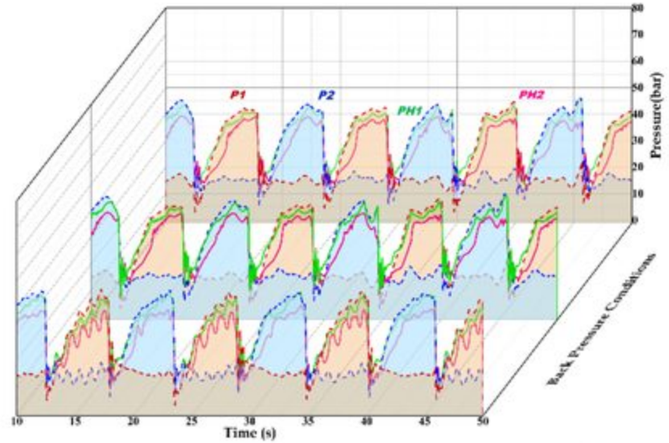
[L1C1] OBP-CDSR



[L1C1] OBP-CDSR

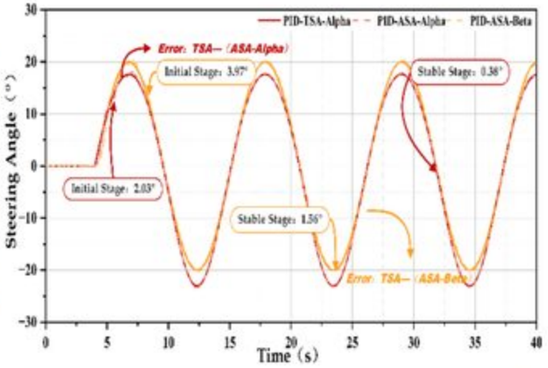


[L1C1] OBP-CDSR

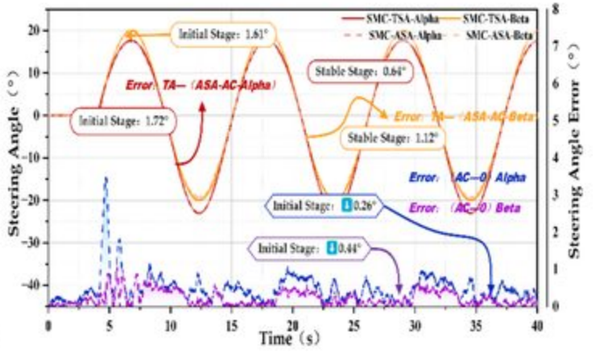


[L3C1] MCBP-CDSR

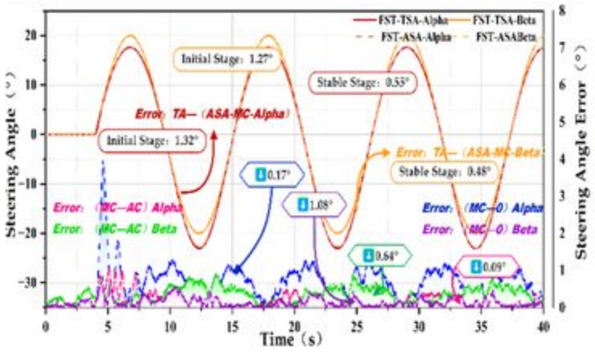
AKMR



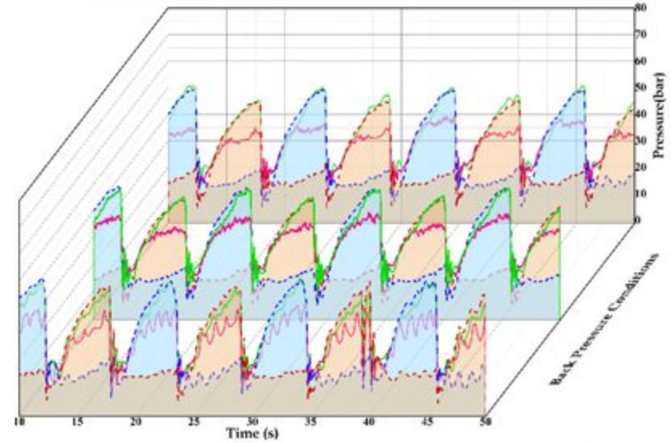
[L1C2] OBP-CDSR



[L1C2] OBP-CDSR



[L1C2] OBP-CDSR



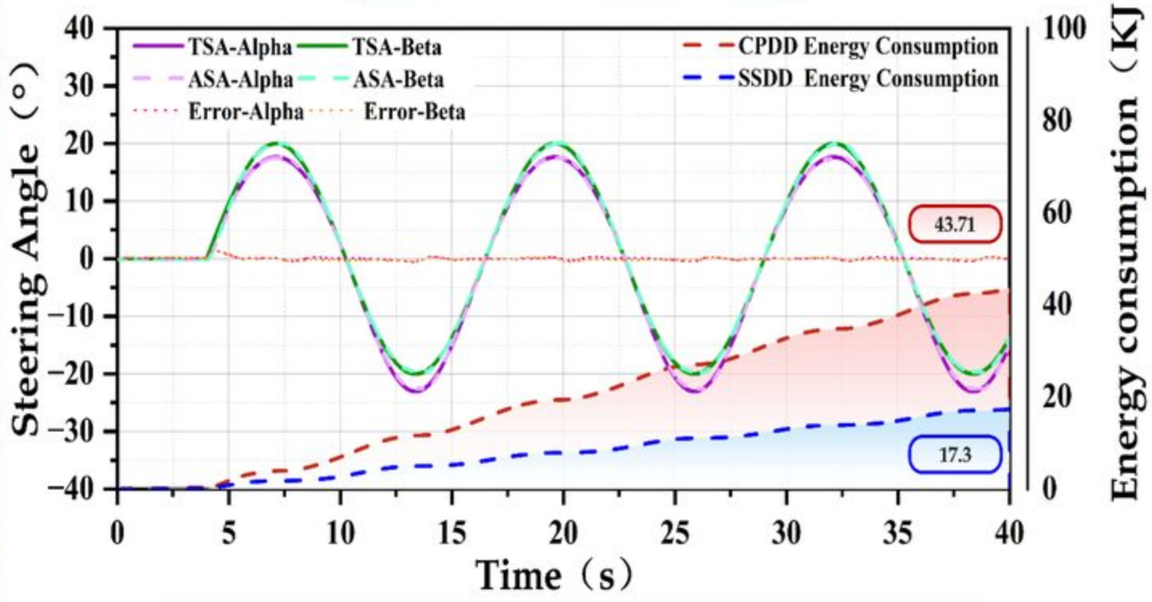
[L3C2] MCBP-CDSR

FST将AKMR模式启动误差控制在1.3°以内，控制精度较PID和SMC分别提高35%和23.5%

【能耗对比】④ 基于参数辨识的能效界定与辨识及非线性鲁棒控制

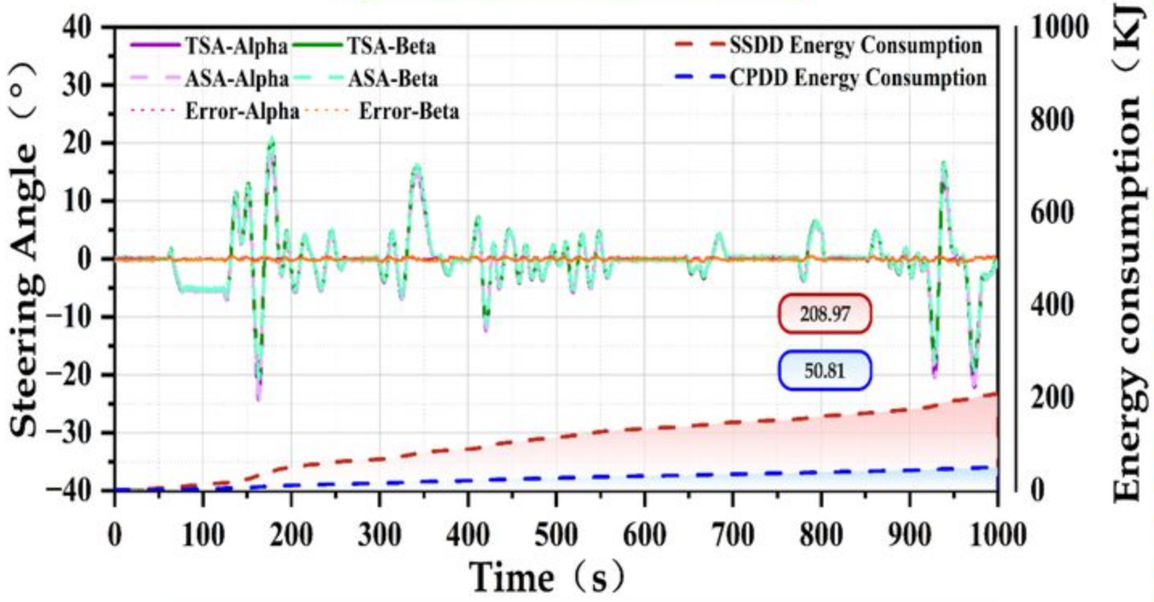
□ 实景环境验证：为验证本方案在真实复杂交通环境下的适应性，模拟重载多轴车载实际驾驶转向工况

固定测试工况



集成FST控制策略下的转角跟踪与能耗

实景环境工况



真实道路行驶模拟下的转向角跟踪与能耗

实景路况测试下EHS-AM总输入能量为 50.81 kJ ， 相较恒压油源CPDD系统节能 75.68%